Lea Duesterwald

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Education

Cornell University, College of Engineering, B.S., Class of 2027 **GPA:** 3.86/4
Major: Computer Science
Relevant Coursework: Principles of Large Scale ML, Machine Learning, Natural Language Processing, AI Reasoning and Decision Making, Computational Genetics and Genomics, Operating Systems, Computer System Organization, Analysis of Algorithms, Object-Oriented Programming, Functional Programming, Linear Algebra, Probability and Statistics, Linguistics
Horace Greeley High School, *Chappaqua, NY*- valedictorian, *summa cum laude*, Class of 2023 **GPA:** 4.10/4

Research Interests

I am interested in understanding how language models learn from their training data, including the influence of data properties, prior learning, and training objectives and architectures on the acquisition of information and capabilities. Developing a deeper understanding of these processes could enable more principled approaches to training, allowing us to more deliberately shape what models learn and the capabilities they acquire.

Academic Research Experience

Student Researcher in the Cornell Relax ML Group Aug. '25. - Present
Advisors: Professor Christopher De Sa (Cornell University) and Professor A. Feder Cooper (Yale University)

- Investigated near-verbatim extraction and memorization in autoregressive language models, contributing to decoding-constrained beam search methods for measuring extractable content beyond exact-match and greedy-decoding approaches.
- Studied how general text predictability affects evidence of memorization, using matched non-training controls to distinguish extractability from information learned from training data.
- Leading research on memorization in diffusion language models, developing likelihood-based evaluation methods to compare memorization across diffusion and autoregressive models and investigating how denoising strategies and training paradigms affect memorization.

Student Researcher at CyLab, Carnegie Mellon University Aug. '24 - Present
Mentor: Professor Norman Sadeh

- Investigated the effectiveness of LLM-generated cybersecurity advice and the impact of prompting on user behavior, leading an in situ user study to evaluate these effects.
- Studied personalization using user background information and conversation history, conducting LLM-based and human evaluations that demonstrated the benefits of conversation-based personalization.

Student Researcher in the Cornell Natural Language Processing Group Feb. - May '25
Advisor: Professor Claire Cardie

- Developed a hybrid GPT-4 and fine-tuned Llama ensemble for numerical claim verification, using verbalized model confidence to select predictions, demonstrating improved classification accuracy over either individual model.

Internship Experience

Intern at Microsoft Research, Redmond WA May - Aug. '25
Mentor: Dr. Suman Nath

- Developed VibeCheck, an MCP server for automated refinement of applications generated by vibe coding agents to deliver users validated applications with fewer bugs.
- Contributed to a multi-team project to improve the efficiency of web agents, including an optimized agent pipeline to work on desktop Windows applications.

Research Experiences for Undergraduates in Software Engineering, Carnegie Mellon University May. - Aug. '24
Mentor: Professor Norman Sadeh

- Developed a Chrome extension-based cybersecurity assistant integrating GPT-4 for conversational question answering.
- Developed and evaluated prompting strategies to improve the effectiveness of LLM-generated cybersecurity advice.

Intern at Northwestern-Argonne Institute of Science and Engineering (NAISE) Jul. - Aug. '22
Mentor: Dr. James Davis (Argonne National Laboratory)

- Developed feature-engineering pipelines and trained machine learning models to predict SARS-CoV-2 PCR cycle-threshold (Ct) values, a proxy for viral load, from genomic sequence data.
- Conducted feature-importance analyses to identify genomic patterns influencing model predictions, including nucleotide k-mers and intrahost sequence variation.

Skills

Programming Languages: Python, Java, JavaScript, C, OCaml, RISC-V Assembly, Bash
ML & Scientific Computing: PyTorch, TensorFlow, NumPy, pandas, scikit-learn, Matplotlib, Hugging Face Transformers, Hugging Face Datasets, SciPy
Tools & Infrastructure: Linux, Git/GitHub, SLURM, SSH, Jupyter, LaTeX, CUDA programming
Certifications: Machine Learning Specialization Certificate, Coursera, Stanford Online (2023)

Presentations

- "Estimating Near-Verbatim Extraction Risk in Language Models with Decoding-Constrained Beam Search." Poster presentation, Conference on Language Modeling (COLM), October 2026.
- "Evaluating the Impact of Personalization in Conversational Cybersecurity Assistants." Poster presentation, HAIPS Workshop at COLM, October 2026.
- "Can a Cybersecurity Question Answering Assistant Help Change User Behavior? An In Situ Study." Oral presentation, Symposium on Usable Security and Privacy (USEC), February 2025. https://youtu.be/AUFyvJMe3Y4?t=3604
- "SARS-CoV-2 genome-based prediction of PCR cycle-threshold values". Research presentation, PATRIC/BV-BRC research meeting, 2022.

Publications

1. A. F. Cooper, M. Swanberg, J. Hayes, L. Duesterwald, C. De Sa, D. E. Ho, M. A. Lemley, and P. Liang. "Extractable Memorization From First Principles." Under review at ICLR 2027. https://dx.doi.org/10.2139/ssrn.7118018
2. A. F. Cooper, M. A. Lemley, C. De Sa, L. Duesterwald, A. Casasola, J. Hayes, K. Lee, D. E. Ho, and P. Liang. "Estimating Near-Verbatim Extraction Risk in Language Models with Decoding-Constrained Beam Search." COLM, 2026. https://doi.org/10.48550/arXiv.2603.24917
3. L. Duesterwald, A. Jain, S. Kochar, and N. Sadeh. "Evaluating the Impact of Personalization in Conversational Cybersecurity Assistants." HAIPS Workshop at COLM, 2026. https://doi.org/10.48550/arXiv.2609.17839
4. L. Duesterwald, A. Arora, and C. Cardie. "CornellNLP at CheckThat! 2025: Hybrid Llama–GPT-4 Ensembles with Confidence Filtering for Numerical Claim Verification." CLEF Working Notes, 2025. https://ceur-ws.org/Vol-4038/paper_68.pdf
5. L. Duesterwald, I. Yang, and N. Sadeh. "Can a Cybersecurity Question Answering Assistant Help Change User Behavior? An In Situ Study." USEC, 2025. https://dx.doi.org/10.14722/usec.2025.23018
6. A. Balaji*, L. Duesterwald*, I. Yang*, A. Priyanshu, C. Alfieri, and N. Sadeh. "Generating Effective Answers to People's Everyday Cybersecurity Questions: An Initial Study." WISE, 2024. https://doi.org/10.1007/978-981-96-0576-7_27 (*equal contribution).
7. L. Duesterwald, M. Nguyen, P. Christensen, S. W. Long, R. J. Olsen, J. M. Musser, and J. J. Davis. "Using Intrahost Single Nucleotide Variant Data to Predict SARS-CoV-2 Detection Cycle Threshold Values." PLOS ONE, 2024. https://doi.org/10.1371/journal.pone.0312686
8. L. Duesterwald, M. Nguyen, P. Christensen, S. W. Long, R. J. Olsen, J. M. Musser, and J. J. Davis. "Using Genome Sequence Data to Predict SARS-CoV-2 Detection Cycle Threshold Values." medRxiv, 2022. https://doi.org/10.1101/2022.11.14.22282297